

Non-Parametric Spatial Correlation Extraction and a Robust Initialization for the Convex-Concave Procedure

From a difference-of-convex MLE to a robust CCP initialization

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corr-solver

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Section 1

Outline

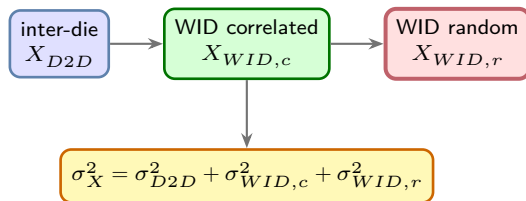
- 1 Outline
- 2 Motivation
- 3 Modeling the Correlation Function
- 4 Problem Formulation
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- 6 Convexity and the CCP
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Section 2

Motivation

- Feature shrink $\Rightarrow$ process variation is a **first-order** determinant of performance and yield
- Corner-based timing verifies at extreme PVT corners $\Rightarrow$ increasingly pessimistic and expensive
- On a 10-inverter chain, worst-case closure cost **+54% area / +44% power** versus statistical design at $\mu + 3\sigma$ [1]
- Statistical timing propagates parameter *distributions* to path delay

Process Variation Decomposition



- $X = \mu_X + X_{D2D} + X_{WID,c} + X_{WID,r}$; the components are independent, so the variances add
- the deterministic layout-dependent part is modeled and removed
- the purely random part together with measurement error becomes the **nugget**

$$\sigma_{DP}^2 = \sum_{i=1}^N \sigma_{D_i}^2 + 2 \sum_{i=1}^N \sum_{j=i+1}^N \text{Cov}(D_i, D_j)$$

- only spatially correlated parameters contribute to the covariance terms
- correlation dominates the path-delay variance as logic depth grows
- the model **must be positive definite**, or the covariance matrix is invalid

Section 3

Modeling the Correlation Function

$$\rho(h) = e^{-\alpha h} \text{ (exp),} \quad e^{-\alpha h^2} \text{ (Gaussian),} \quad \text{Matérn}$$

- positive definite **by construction**, parameterized by a length scale [2], [3]
- but they **presuppose the shape**, and their likelihood is **non-convex**
- for a new process the correlation may even be **non-monotone**

$$\rho(h) = \sum_i p_i \Psi_i(h)$$

- polynomial and B-spline bases commit to **no particular shape**
- they can represent non-monotone correlation
- the price: a constrained, higher-dimensional fit, and positive definiteness must be enforced separately

- purely random intra-die variation **and** measurement error give white noise of variance τ^2
- this introduces a discontinuity at the origin:

$$\tilde{R}(h) = \begin{cases} 1, & h = 0 \\ \frac{\sigma^2}{\sigma^2 + \tau^2} R(h), & h > 0 \end{cases} \quad \tilde{R}(\kappa, \psi) = R(\psi) + \kappa I, \quad \kappa = \frac{\tau^2}{\sigma^2}$$

- ignoring the nugget severely distorts the recovered correlation function

- ① basis-expansion formulation of the covariance; LSQ and MLE fit by a **cutting-plane** method
- ② the MLE objective is a **difference of convex** functions $\Rightarrow$ not globally convex
- ③ solved with the **convex-concave procedure (CCP)**; its fixed points are stationary points
- ④ **robust initialization**: free-MLE covariance plus a nearest-PD repair – the main result
- ⑤ validated on polynomial & clamped B-spline bases, isotropic & anisotropic fields, in Python / C++ / Rust

Section 4

Problem Formulation

$$Y = \frac{1}{M} \sum_{m=1}^M y_m y_m^\top$$

- $M \geq n$: Y is positive definite; $M < n$: rank-deficient and **singular**
- “more samples is easier”: Y approaches the true covariance and the problem becomes better conditioned

$$\rho(h) = \sum_i p_i \Psi_i(h), \quad \Omega(p) = \sum_{k=1}^m p_k F_k, \quad (F_k)_{ij} = \Psi_k(\|s_i - s_j\|)$$

- $\Omega(p)$ is **affine** in the coefficients p – inviting a convex program
- but the MLE objective contains $\Omega(p)^{-1}$, and a matrix inverse is not convex

$$\min_{p, \kappa} \|\Omega(p) + \kappa I - Y\|_F \quad \text{s.t.} \quad \Omega(p) \succeq 0, \quad \kappa \geq 0$$

- a **convex** problem; the nugget κI gives strict positive definiteness
- it targets the *shape* of the sample covariance and places no upper bound on Ω

$$\log L = -\frac{MN}{2} \log \sigma^2 - \frac{M}{2} \log \det \tilde{R} - \frac{1}{2\sigma^2} \sum_m z_m^{*\top} \tilde{R}^{-1} z_m^*$$

- concentrating out σ^2 gives $\log L_0(\kappa, \psi) = -\log \det \tilde{R} - N \log(\text{tr}(Y \tilde{R}^{-1}))$
- pooling **all** M **chips** into one likelihood yields a single parameter set [4]
- steps: center z^* ; form Y ; minimize $g = \log \det \tilde{R} + N \log \text{tr}(Y \tilde{R}^{-1})$; then $\hat{\sigma}^2 = \frac{1}{N} \text{tr}(Y \tilde{R}^{-1})$
- $\log \det$ is evaluated via LU, because a direct determinant underflows

Section 5

Non-Parametric Bases

$$\varphi_k(d) = d^k, \quad k = 0, \dots, m-1$$

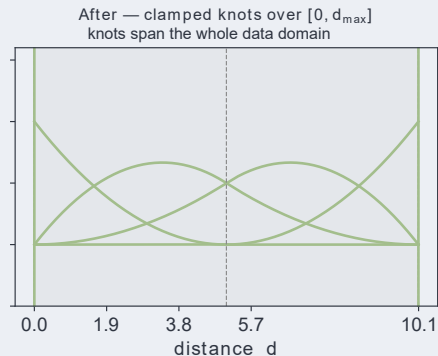
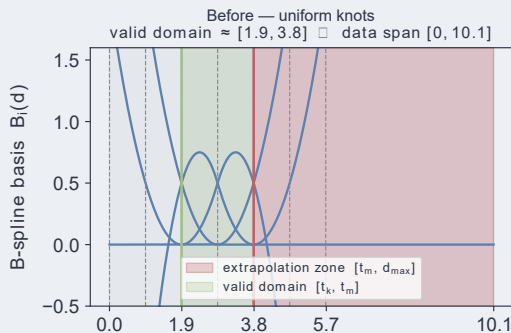
- every basis function has **global support**, so the design columns correlate as m grows
- $\text{cond}(A)$ climbs from 1.2×10^1 at $m = 2$ to 3.5×10^{10} at $m = 10$
- nine orders of magnitude on the *same* problem

$$K(d) = \sum_{i=1}^m c_i B_i(d), \quad B_i \geq 0, \quad \sum_i B_i = 1$$

- local support of roughly three knot spans
- the knots must span the **actual data domain** $[0, d_{\max}]$
- legacy uniform knots extrapolate over most of the domain $\Rightarrow$ numerically meaningless
- clamped knots repeat the end knots $k + 1$ times

Clamped Knots

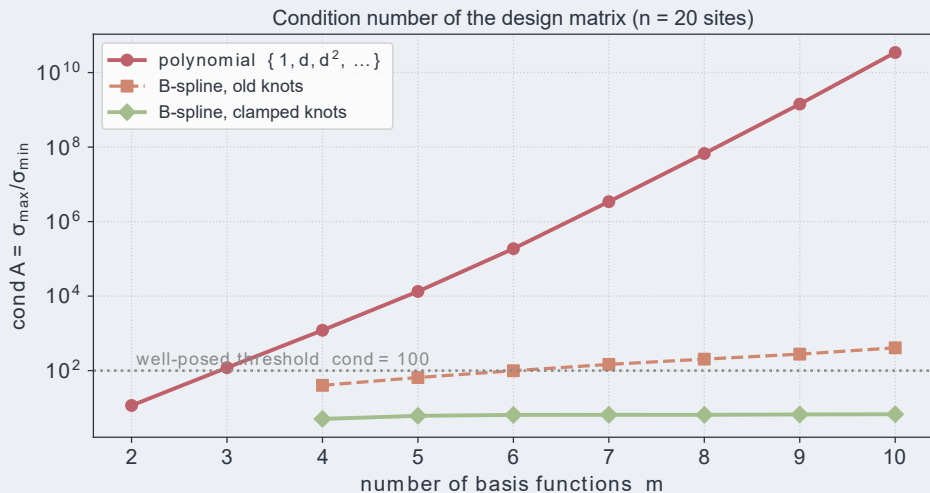
$m = 4$ quadratic B-splines: where can the basis actually live?



m	polynomial	B-spline (clamped)	B-spline (legacy)
4	1.21×10^3	5.05	40.26
6	1.87×10^5	6.48	99.40
8	6.72×10^7	6.51	203.57
10	3.45×10^{10}	6.75	409.47

- the clamped B-spline stays near 5–7 across $m = 4, \dots, 10$, **at no cost in fidelity**
- a non-increasing coefficient sequence gives a monotone spline; apply it to *exactly* the control coefficients

Design-Matrix Conditioning



Section 6

Convexity and the CCP

The MLE Objective Is Not Convex

$$f(\Omega) = \underbrace{\text{tr}(\Omega^{-1}Y)}_{\text{convex}} + \underbrace{\log \det \Omega}_{\text{concave}}$$

- a **difference of convex** (DC) functions, hence in general neither convex nor concave
- the ellipsoid method **cannot** be applied directly

The Convex-Concave Procedure

- $\log \det \Omega$ is concave, so its tangent is a **global upper bound**:

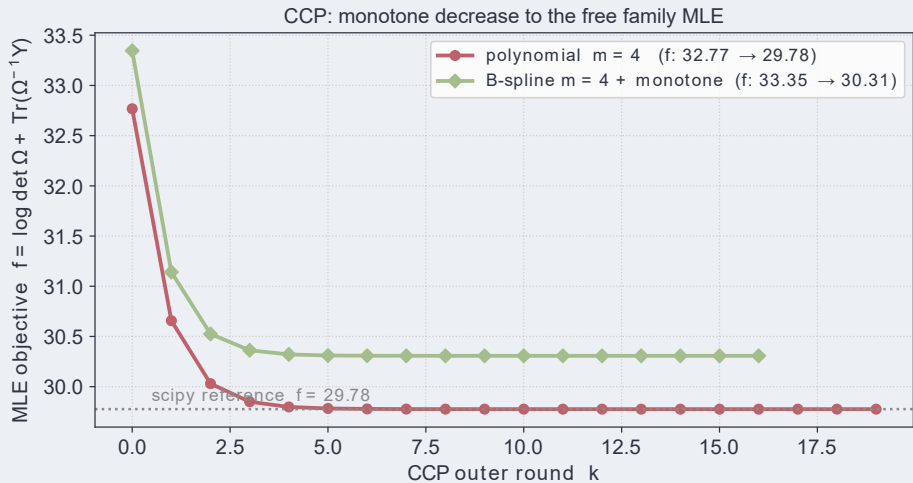
$$\log \det \Omega \leq \log \det \Omega_k + \text{tr}(\Omega_k^{-1}(\Omega - \Omega_k))$$

- dropping constants gives the convex surrogate

$$S_k(\Omega) = \text{tr}(\Omega^{-1}Y) + \text{tr}(M_k\Omega), \quad M_k = \Omega_k^{-1}$$

- monotone decrease: $f(\Omega_{k+1}) \leq S_k(\Omega_{k+1}) \leq S_k(\Omega_k) = f(\Omega_k)$
- a fixed point is a **stationary point** of the true MLE

CCP Convergence



$$\min_p \operatorname{tr}(\Omega(p)^{-1}Y) + \operatorname{tr}(M_k \Omega(p)) \quad \text{s.t.} \quad \Omega(p) = \sum_i p_i F_i \succeq 0$$

- a **linear matrix inequality** in p : only m variables, independent of n
- solved by the **ellipsoid method** [5], which needs only a separation oracle

- at p_c the separation oracle returns a **cut** (g, β) with $g^\top(p - p_c) + \beta \leq 0$
- the deep-cut update:

$$p_c^+ = p_c - \frac{\rho}{\tau^2} \tilde{g}, \quad P^+ = \delta \left(P - \frac{\sigma}{\tau^2} \tilde{g} \tilde{g}^\top \right)$$

- $\text{LDL}^\top$ factor of $\Omega(p_c)$: positive pivots $\Rightarrow$ feasible; failure yields a witness cut
- **lazy**: the factorization stops at the first failing pivot, $O(p^3)$
- the volume shrinks $\approx e^{-1/(2m)}$ per step $\Rightarrow O(m^2 \log(1/\varepsilon))$ iterations

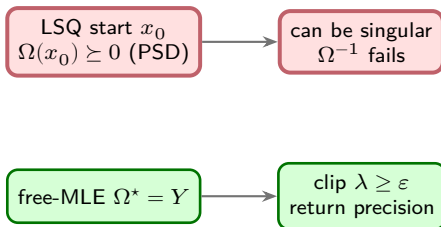
- ① $M \leftarrow \text{mle_corr_mtx}(Y)$ — the robust initialization of the next section
- ② **repeat** until the objective stalls:
- ③ solve the convex subproblem with $M \Rightarrow x'$
- ④ if x' is missing or stalled, stop
- ⑤ $x \leftarrow x'$; $M \leftarrow \Omega(x)^{-1}$

Section 7

Robust Initialization

The Problem: PSD Is Not PD

- the old start used the **LSQ** solution x_0 and set $M_0 = \Omega(x_0)^{-1}$
- but LSQ enforces only $\Omega(x_0) \succeq 0$ – the *boundary* is allowed
- the CCP majorization needs $\Omega_0 \succ 0$ for Ω_0^{-1} and $\log \det \Omega_0$
- concrete failure: polynomial basis $F_0 = \mathbf{1}$, $x_0 = [4, 0, 0, 0] \Rightarrow \Omega(x_0) = 4 \cdot \mathbf{1}$, rank one $\Rightarrow$ **error**



$$\min_{\Omega \succ 0} \log \det \Omega + \text{tr}(\Omega^{-1}Y)$$

- set the gradient to zero: $\Omega^{-1} - Y = 0 \Rightarrow \Omega^* = Y$
- the free-MLE covariance is simply the **sample covariance**
- a natural initialization: the exact minimizer of the objective *without* the family constraint

- the free problem is unbounded when Y is not positive definite
- repair to the nearest PD matrix in Frobenius norm by **spectral clipping** [6]:

$$\Omega_0 = V \operatorname{diag}(\max(\lambda, \varepsilon)) V^\top$$

- positive definite by construction $\Rightarrow$ **always invertible**

- the CCP never needs Ω_0 itself – only its inverse
- reciprocate the clipped eigenvalues:

$$\Omega_0^{-1} = V \operatorname{diag}\left(\frac{1}{\max(\lambda, \varepsilon)}\right) V^\top$$

- one symmetric eigendecomposition, **no external SDP solver**
- if $Y \succ 0$ the floor is inactive and the initializer is exactly Y^{-1}

- ① symmetrize: $Y \leftarrow (Y + Y^\top)/2$
- ② eigendecompose $Y = V \text{diag}(w) V^\top$ (ascending)
- ③ $\delta \leftarrow \varepsilon \cdot \max(w_{\max}, 1)$
- ④ $w \leftarrow \max(w, \delta)$
- ⑤ **return** $V \text{diag}(1/w) V^\top$, i.e. Ω_0^{-1} directly

Section 8

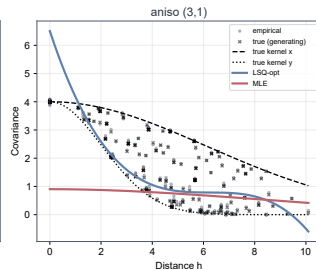
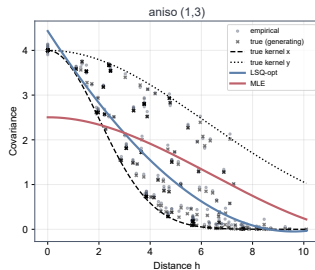
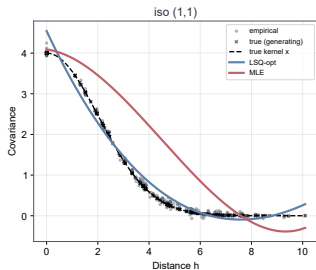
Numerical Experiments

- Halton low-discrepancy sites with a known generating kernel, isolating the *algorithm*
- $n = 20$ sites, $N = 3000$ samples; isotropic $K(d) = 4 e^{-0.12d^2}$
- **Cholesky generator** [7]: $V = \sigma^2 K + \tau^2 I = LL^\top$; $y_m = Lx_m$, $x_m \sim \mathcal{N}(0, I)$;
 $Y = \frac{1}{M} \sum_m y_m y_m^\top$
- synthetic data establishes *algorithmic* behavior, not production readiness

problem	LSQ	MLE	CCP
iso (1, 1)	0.128	0.705	0.293
aniso (1, 3)	0.334	0.451	0.451
aniso (3, 1)	0.476	0.716	0.683

- LSQ tracks the shape; the MLE oracle is pinned to its bound and near-singular; the CCP recovers the free family MLE
- the MLE gap does **not** close as $N \rightarrow \infty \Rightarrow$ structural **model misspecification**

Fitted Correlation Functions



- the nugget-aware **MLEnug** keeps $\text{err}(R(h)) < 2.3\%$ in every regime
- MLEsim** (ignores the nugget) reaches 78% error; MLEnug dominates the least-squares **RESCF** *and* runs faster

scheme	method	$\text{err}(\sigma^2)$	$\text{err}(R(h))$
uniform	RESCF (LSE)	0.95–7.31%	0.75–4.26%
uniform	MLEsim	10.98–99.18%	5.44–55.72%
uniform	MLEnug	0.30–4.12%	0.27–2.27%
Monte Carlo	RESCF (LSE)	0.91–2.32%	1.06–2.33%
Monte Carlo	MLEsim	7.58–86.13%	42.48–77.92%
Monte Carlo	MLEnug	0.32–1.67%	0.29–1.39%

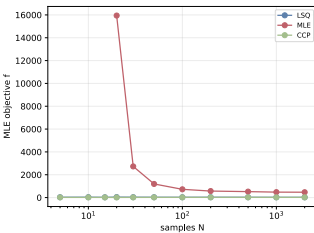
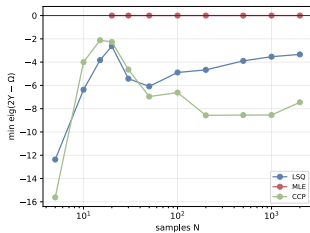
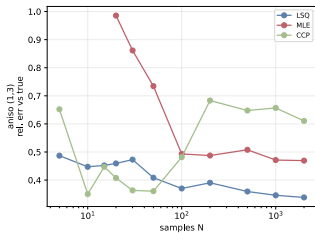
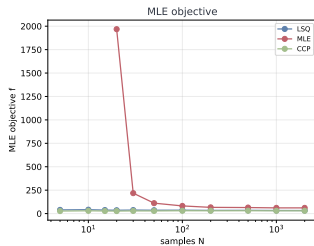
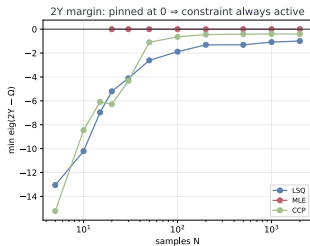
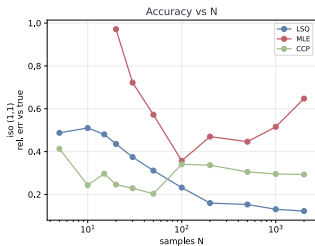
Robust Initialization: Same Optimum

	LSQ warm start (old)	free-MLE init (new)
CCP objective f	29.775870	29.775870
coefficient rel. diff	—	1.9×10^{-4}
linearization rounds	16	18
singular $\Omega(x_0)$	error	works

- identical optimum; the new initializer adds **two** rounds
- it tolerates a singular start that previously raised an error
- the initialization changes the **optimization path**, not the answer

- $M \geq n$: the error falls like $1/\sqrt{M}$ and flattens beyond ≈ 200 chips
- $M < n$: Y is rank-deficient and the MLE oracle is infeasible
- the monotone B-spline CCP stays **well-posed** even at the rank-deficient $N = 1$

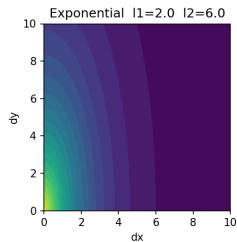
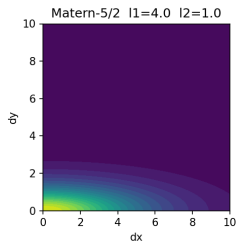
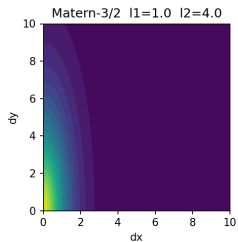
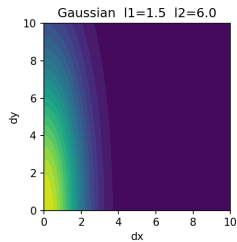
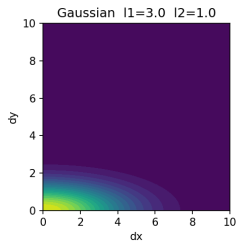
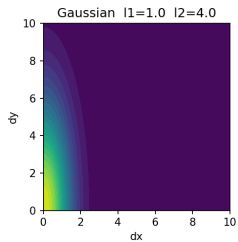
Accuracy vs Sample Size



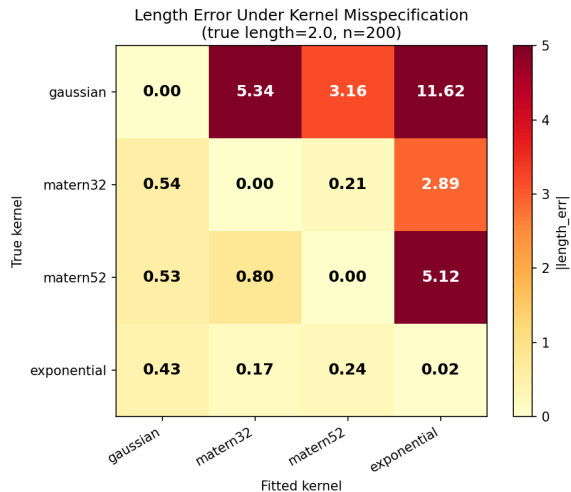
- anisotropic length scales recovered to $\approx 1\%$ (Gaussian / Matérn), a few % (exponential)
- misspecification is **not** cured by more data: fitting a smooth kernel with a rough one inflates ℓ many-fold
- no parametric kernel is a safe default $\Rightarrow$ **validate** (fit several, compare held-out likelihood), don't assume

Anisotropic Correlation Surfaces

Anisotropic Correlation Surfaces $C(dx, dy)$



Kernel Misspecification



- the algorithms were reimplemented in **Python, C++, and Rust**
- conditioning agrees to nine significant digits; the reference value at $m = 10$ was adjudicated at sixty digits
- two bugs found and fixed: a triangular inverse that assumed symmetry, and monotonicity index arithmetic that assumed a trailing objective variable
- all three implementations now agree on every reported quantity

Section 9

Discussion and Conclusion

- **synthetic only**: Cholesky fields with analytical kernels; no non-stationarity, non-Gaussian noise, or lithographic signatures
- objective alignment: LSQ tracks *shape*, the MLE gives a *calibrated likelihood*
- boundary protection guarantees positive definiteness only for the **first** round
- scalability: an $O(n^3)$ factorization per query limits dense full-chip use ($n > 10^3$)
- the site-wise LMI certifies PD only at the sites, not on the continuum (needs Bochner)

When Is Non-Parametric CCP Warranted?

- family **known** $\Rightarrow$ fit the parametric kernel (simpler, more accurate)
- shape **unknown** or **non-monotone** $\Rightarrow$ clamped monotone B-spline plus CCP
- a misspecified kernel is simply wrong, however good its residual looks

- non-parametric extraction = basis expansion + cutting-plane fit (LSQ / MLE)
- the MLE objective is a difference of convex functions, solved by the CCP
- **robust init**: free-MLE $\Omega^* = Y$, repaired by spectral clipping, precision returned directly
- always positive definite, removes the singular failure mode, same optimum at +2 iterations
- validated on polynomial / B-spline bases, isotropic / anisotropic fields, in Python / C++ / Rust

Questions?

code: <https://github.com/luk036/corr-solver>

- paper: `paper/spatial.md`

- [1] V. Champac and J. García Gervacio, *Timing performance of nanometer digital circuits under process variations*, vol. 39. in *Frontiers in electronic testing*, vol. 39. Springer, 2018.
- [2] B. Matérn, *Spatial variation*, 2nd ed. Springer, 1986.
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- [6] N. J. Higham, "Computing a nearest symmetric positive semidefinite matrix," *Linear Algebra and its Applications*, vol. 103, pp. 103–118, 1988.
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